

NEUROSCI 366

Lecture 24

从奖励中学习 / Learning from Rewards

Source-aligned · Static · Searchable · Printable

Source files

- Lecture24-LearningFromRewards.pdf

Production version: 2026-08-22

Reader profile: neuroscience undergraduate education; mathematics scaffolded from high-school algebra/trigonometry to the formal computational-neuroscience level.

This companion preserves source-page order, distinguishes source material from necessary scaffolding and extensions, and places hints/answers after the lesson. It is a static PDF: no forms, JavaScript, hidden-answer widgets, passwords, or DRM.

Contents

1	How to use this companion	2
2	Learning objectives and prerequisite dependency map	2
3	Five-minute prerequisite diagnostic	2
4	Source-aligned lesson / 按原笔记页序	3
4.1	Lecture24-LearningFromRewards.pdf · p. 1	3
4.2	Lecture24-LearningFromRewards.pdf · p. 2	4
4.3	Lecture24-LearningFromRewards.pdf · p. 3	5
4.4	Lecture24-LearningFromRewards.pdf · p. 4	6
4.5	Lecture24-LearningFromRewards.pdf · p. 5	7
4.6	Lecture24-LearningFromRewards.pdf · p. 6	8
5	补全推导与数学支架 / Derivations	9
5.1	reward-baseline three-factor rule	9
5.2	matching condition	10
5.3	Rescorla-Wagner 与 blocking	11
5.4	TD error、temporal features 与 Q	12
6	Cross-page synthesis / 跨页因果与数学链	13
7	Worked examples and transfer	13
7.1	dopamine RPE 三模式	13
7.2	TD 数值	13
8	Formula and notation sheet	15
9	Bilingual glossary	15
10	Common traps, assumptions, and limitations	16
11	Cumulative Knowledge Check	17
12	Hint Bank	19
13	Complete Answer Key with reasoning	21
14	Source-page concordance and coverage audit	22

15 Errata, uncertainty log, and external endnotes	22
15.1 Errata / source cautions	22
15.2 Uncertainty log	22
15.3 External endnotes	22
15.4 Final self-audit	23

1 How to use this companion

1. 先用 5-minute diagnostic 暴露前置缺口；不要立即翻答案。
2. 按原笔记页序阅读 source-page units：先看原页，再读 clean reconstruction 与补全逻辑。
3. 遇到公式按“问题 → 符号/shape/units → assumptions → 一行一步推导 → intuition → biological meaning → sanity check → failure”阅读。
4. 完成 Cumulative Knowledge Check；只在需要时依次打开 Hint 1、2、3。
5. 至少隔一个 page break 后再核对 Answer Key，并记录错误类型：concept、symbol、algebra、assumption、correlation/causation。

Source hierarchy. 原笔记是内容权威；本书中的“【必要前置】/【补全推导】/【跨课连接】/【延伸】”是为了补齐数学或迁移，不替代源材料。无法确认的内容必须进入 Uncertainty Log。

2 Learning objectives and prerequisite dependency map

- 定义并区分：stimulus/action/reward。
- 从第一原则推导：three-factor plasticity。
- 解释几何/计算直觉：reward baseline/covariance。
- 连接生物学解释：matching law。
- 诊断 assumptions 与 failure modes：Rescorla-Wagner。
- 把概念迁移到新神经科学问题：TD/RPE。

Dependency map

stimulus/action/reward → three-factor plasticity → reward baseline/covariance → matching law → Rescorla-Wagner → TD/RPE

Core question

reward-gated plasticity 如何避免无条件增权并产生 matching, value prediction 与 temporal-difference error 又如何解释 blocking 与 dopamine responses?

3 Five-minute prerequisite diagnostic

1. naive reward-Hebb 为什么 over-strengthen? (卡住时先看 Source-page unit 1, 再看补全推导。)
2. 减 reward baseline 后 expected update 变什么? (卡住时先看 Source-page unit 2, 再看补全推导。)
3. matching 的 marginal-return condition? (卡住时先看 Source-page unit 3, 再看补全推导。)
4. matching law 的 ratio? (卡住时先看 Source-page unit 4, 再看补全推导。)
5. Rescorla-Wagner error? (卡住时先看 Source-page unit 5, 再看补全推导。)

Scoring guide: 4–5 confident answers → proceed; 2–3 → use the prerequisite labels; 0–1 → read the formula/notation sheet once before the source-aligned lesson.

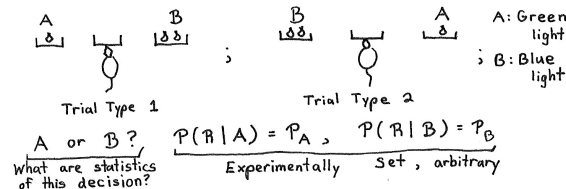
4 Source-aligned lesson / 按原笔记页序

4.1 Lecture24-LearningFromRewards.pdf • p. 1

Learned Decisions:

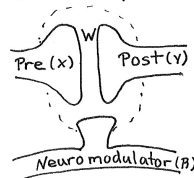
Many real world decisions require learning.

Foraging: Am I more likely to find food I want at Trader Joe's or Jewel? Is a rat more likely to find food in a blue or green garbage bin?



This behavior requires learned associations between stimulus (color of light) and reward (amount of water).

Three-factor plasticity



$$\Delta W = \Delta W(x, y, R)$$

R: Reward

Neuromodulator canonically associated with reward learning is Dopamine.

Other canonical learned decisions:

- Navigation: How do I find the water source?
- Economic valuation: How much will I pay for brand-name shampoo?

Original-note anchor. [Source: Lecture24-LearningFromRewards.pdf, p. 1]

Clean reconstruction. Page 1: learned decisions, stimulus-reward association, three-factor plasticity, dopamine.

What the note is saying. 本页的中心对象是 stimulus/action/reward 与 three-factor plasticity。原笔记将它们放进以下链条: learned decisions, stimulus-reward association, three-factor plasticity, dopamine。

Why it follows. 用 foraging 示例解释 stimulus/action/reward association 与 three-factor plasticity; 明确 dopamine 是 canonical candidate third factor 而非所有 reward learning 的唯一机制。其逻辑后果是: 重绘 dopamine RPE 的 early reward、cue transfer、omitted reward 三种模式, 并区分 RPE、value 和 salience alternative。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 按时间线区分 stimulus/task、recording/manipulation、measured quantity 与 inferred model variable; 结果相关性不能直接替代机制检验。

Stop & Predict

naive reward-Hebb 为什么 over-strengthen?

4.2 Lecture24-LearningFromRewards.pdf • p. 2

Covariance-based plasticity:

Dopamine speeds up plasticity, and the simplest models treat it as a gain parameter.

$$\Delta W = \underbrace{S(x, y)}_{\text{e.g. Hebbian}} R$$

like a learning rate

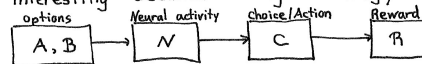
This can over-strengthen the synapse when reward keeps coming. Stop learning when reward is expected:

$$\Delta W = S(x, y)(R - \langle R \rangle)$$

Note that

$$\langle \Delta W \rangle = \langle S(x, y)(R - \langle R \rangle) \rangle = \text{Cov}(\underbrace{S(x, y)}_{\text{"N"}}, R)$$

so this is again a "covariance rule." Loewenstein & Seung showed that such plasticity rules lead to an interesting decision-making strategy.



$$\langle \Delta W \rangle = 0 \Leftrightarrow \text{Cov}(N, R) = 0 \Leftrightarrow \langle R \rangle_{N|C=A} = \langle R \rangle_{N|C=B}$$

The first statement follows directly from plasticity rule.

The second statement follows from

$$\begin{aligned} \text{Cov}(N, R) &= \langle (N - \langle N \rangle) R \rangle = \langle \langle (N - \langle N \rangle) R \rangle_{N|C} \rangle_C \\ &= P(C=A) \langle \delta N \rangle_{N|C=A} \langle R \rangle_{R|C=A} + P(C=B) \langle \delta N \rangle_{N|C=B} \langle R \rangle_{R|C=B} \\ &\quad \text{Reward depends on N only through choice} \\ &= P(C=A) \langle \delta N \rangle_{N|C=A} (\langle R \rangle_{R|C=A} - \langle R \rangle_{R|C=B}) \end{aligned}$$

The condition follows if we assume that the mean neural activity depends on choice (i.e. $\langle \delta N \rangle_{N|C=A} \neq 0$).

Original-note anchor. [Source: Lecture24-LearningFromRewards.pdf, p. 2]

Clean reconstruction. Page 2: covariance-based reward plasticity, reward-baseline subtraction, matching condition derivation.

What the note is saying. 本页的中心对象是 reward baseline/covariance 与 matching law。原笔记将它们放进以下链条：covariance-based reward plasticity、reward-baseline subtraction、matching condition derivation。

Why it follows. 逐步推导 matching condition: choice-conditioned expected rewards 相等如何导出 choice ratio $\approx$ reward ratio; 说明 assumptions。其逻辑后果是：从 naive reward-gated Hebb rule 引出 over-strengthening, 再推导 reward-baseline subtraction 和 covariance form。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 把图与矩阵 shape 对齐：轴代表变量或 mode, 椭圆方向对应 eigenvectors, 轴长/曲率对应 eigenvalues; 不要把统计轴自动解释为因果通路。

Stop & Predict

減 reward baseline 后 expected update 变什么?

4.3 Lecture24-LearningFromRewards.pdf • p. 3

Matching behavior:

The condition that

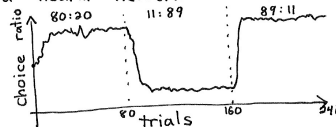
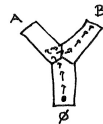
$$\langle R \rangle_{R|C=A} = \langle R \rangle_{R|C=B}$$

is known as "matching", because it implies that the animal matches its choice ratio to the reward ratio

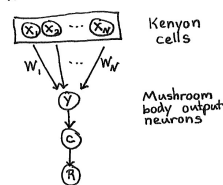
$$\frac{R_A}{C_A} = \frac{R_B}{C_B} \Rightarrow \frac{C_A}{C_B} = \frac{R_A}{R_B}$$

R_A/B : Total rewards from options
 C_A/B : Total choices (by Herrnstein)

Matching was first described in pigeons and has since been found in humans, monkeys, mice, flies, and more. My collaborators and I discovered matching in flies and built a neural network model for it.



Computing choice and reward ratios over many blocks and flies revealed matching. Our neural network model was of the mushroom body.



$$Y = \sum_{i=1}^N W_i X_i$$

$$P(C = \text{"Approach"}) = \frac{e^{-g(Y-\alpha)}}{1 + e^{-g(Y-\alpha)}}$$

g : controls stochasticity of choice
 α : biases choice

$$\Delta W_i = -\sum X_i (R - \langle R \rangle)$$

Model needed $\langle R \rangle$ term to generate matching and fit sequential decisions of flies.

Original-note anchor. [Source: Lecture24-LearningFromRewards.pdf, p. 3]

Clean reconstruction. Page 3: matching law, fly foraging neural model.

What the note is saying. 本页的中心对象是 Rescorla-Wagner 与 TD/RPE。原笔记将它们放进以下链条: matching law、fly foraging neural model。

Why it follows. 重构 fly mushroom-body model、stochastic choice 与 baseline reward estimate; 解释 matching 的行为与神经实现层次。其逻辑后果是: 逐步推导 matching condition: choice-conditioned expected rewards 相等如何导出 choice ratio $\approx$ reward ratio; 说明 assumptions。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系, 再问它表达的是定义、推导、模型预测还是经验观察; 示意图不提供未标注的定量精度。

Stop & Predict

matching 的 marginal-return condition?

4.4 Lecture24-LearningFromRewards.pdf • p. 4

Predicting rewards:

Many models posit that animals predict rewards through a "value function." In the simplest case,

$$V = \sum_{i=1}^N \underbrace{w_i}_{\text{weighting coefficients}} \underbrace{x_i}_{\text{stimuli}}$$

The value is meant to predict rewards. Minimizing $(R-V)^2$ with gradient descent gives the Rescorla-Wagner rule.

$$\Delta w_i = \alpha x_i (R - V)$$

This rule predicts many results in classical conditioning. For example, Pavlovian associations occur when X predicts R through V . Blocking occurs because if S_1 predicts R , then there is no learning signal for the new associations of $S_1 + S_2 \rightarrow R$, so $S_2 \rightarrow \emptyset$.

Temporal difference (TD) learning is designed to predict rewards that require time for delivery. For instance, to navigate animals learn a sequence of actions that eventually lead to reward, and each decision is usually not accompanied by immediate reward. Want to learn an estimate of total future reward

$$\begin{aligned} R(t) &= R(t) + \gamma R(t+1) + \gamma^2 R(t+2) + \dots \\ &= R(t) + \underbrace{\gamma}_{\text{Temporal discounting}} R(t+1) \end{aligned}$$

An ideal value function would report

$$V(t) = \langle R(t) \rangle$$

$$R(t) = V(t) - \gamma V(t+1)$$

So the relevant error signal is

$$\delta(t) = R(t) - V(t) + \gamma V(t+1)$$

Original-note anchor. [Source: Lecture24-LearningFromRewards.pdf, p. 4]

Clean reconstruction. Page 4: value prediction, Rescorla-Wagner, blocking, temporal difference/discounting.

What the note is saying. 本页的中心对象是 Q/action selection 与 stimulus/action/reward。原笔记将它们放进以下链条: value prediction、Rescorla-Wagner、blocking、temporal difference/discounting。

Why it follows. 从 linear value prediction 和 squared prediction error 推导 Rescorla-Wagner; 用 blocking 解释 zero prediction error。其逻辑后果是: 解释 temporal features/state representation、Q value 与 greedy action selection。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系, 再问它表达的是定义、推导、模型预测还是经验观察; 示意图不提供未标注的定量精度。

Stop & Predict

matching law 的 ratio?

4.5 Lecture24-LearningFromRewards.pdf • p. 5

We again predict value through stimuli,

$$V(t) = \underbrace{\sum_{t'=0}^t \sum_{i=1}^N W_i(t') x_i(t-t')}_{\text{Discrete time convolution}}$$

and update weights as

$$\Delta W_i(t') = \gamma \delta(t) x_i(t-t')$$

Note that V depends on stimuli and state (e.g., how long a stimulus has been present). Denote this combination as Δ_t . It's often convenient to seek a value function that depends also on chosen actions.

Q-learning: Seek to learn

$$Q(\underbrace{\Delta_t}_{\text{state}}, \underbrace{a_t}_{\text{action}}; t) = \langle R(t) \rangle_{R(t), R(t+1), \dots | \Delta_t = \Delta, a_t = a}$$

Q is called the state-action value function, or simply the "Q-value."

Original-note anchor. [Source: Lecture24-LearningFromRewards.pdf, p. 5]

Clean reconstruction. Page 5: temporally extended features, state-action value/Q.

What the note is saying. 本页的中心对象是 three-factor plasticity 与 reward baseline/covariance。原笔记将它们放进以下链条: temporally extended features、state-action value/Q。

Why it follows. 从 linear value prediction 和 squared prediction error 推导 Rescorla-Wagner; 用 blocking 解释 zero prediction error。其逻辑后果是: 重构 fly mushroom-body model、stochastic choice 与 baseline reward estimate; 解释 matching 的行为与神经实现层次。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系, 再问它表达的是定义、推导、模型预测还是经验观察; 示意图不提供未标注的定量精度。

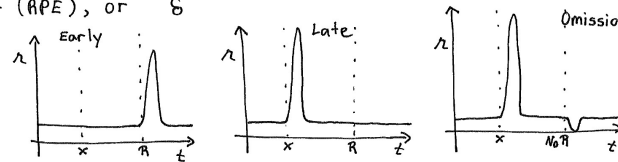
Stop & Predict

Rescorla-Wagner error?

4.6 Lecture24-LearningFromRewards.pdf • p. 6

Neurobiology of RPE & Value:

Schultz, Dayan, and Montague famously argued that dopamine neuron activity resembles the reward prediction error (RPE), or δ



These dopamine neurons project to striatum. Subsequent work has found neural activity correlated to $V(t)$ and/or $Q(\Delta, a; t)$ in striatum. Such signals could support action selection. For instance

$$a_t = \arg \max_a Q(\Delta, a; t)$$

Original-note anchor. [Source: Lecture24-LearningFromRewards.pdf, p. 6]

Clean reconstruction. Page 6: dopamine reward-prediction error patterns, value/Q signals, action selection.

What the note is saying. 本页的中心对象是 matching law 与 Rescorla-Wagner。原笔记将它们放进以下链条：dopamine reward-prediction error patterns、value/Q signals、action selection。

Why it follows. 解释 temporal features/state representation、Q value 与 greedy action selection。其逻辑后果是：从 delayed rewards 推导 discounted return、value function 和 TD error $\delta = R - V + \gamma V_{\text{next}}$ 。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系，再问它表达的是定义、推导、模型预测还是经验观察；示意图不提供未标注的定量精度。

Stop & Predict

blocking 为什么发生?

5 补全推导与数学支架 / Derivations

5.1 reward-baseline three-factor rule

【必要前置】 为什么 $\Delta w = \text{pre} \times \text{post} \times R$ 会 over-strengthen?

$$\Delta w = \eta e(x, y) R$$

$$\Delta w = \eta e(x, y) (R - \bar{R})$$

$$E[\Delta w] = \eta \text{Cov}(e, R)$$

1. eligibility/coactivity e 在行为与 reward 发生时为正。
2. 若 R 总为正, naive rule 平均持续 potentiates。
3. 减去 expected reward 使普通/已预测 reward 的 mean update 为零。
4. 只有 coactivity 与 above-baseline reward 共变时净增强。

Geometric/computational intuition. 学习的是“比预期好/坏”, 而不是 reward 的绝对存在。

Biological interpretation / failure condition. baseline 估计错误、delayed credit assignment 与 nonstationary rewards 需更复杂 mechanisms。

Micro-check

matching law 的 ratio?

5.2 matching condition

【补全推导】 stationary learning 如何导出 choice ratio $\approx$ reward ratio?

$$E[\Delta w] = 0 \Rightarrow \text{Cov}(N, R) = 0$$

$$E[R \mid C = A] = E[R \mid C = B]$$

$$\frac{C_A}{C_B} \approx \frac{R_A}{R_B}$$

1. 假设 neural activity N 的 conditional mean 随 choice 改变。
2. covariance=0 可分解为 choice-conditioned reward means 的差。
3. stationarity 要求两 choice 的 obtained reward rate 相等，否则权重继续改变。
4. 在 baiting/foraging assumptions 下，这等价于 choice fractions 匹配 reward fractions。

Geometric/computational intuition. agent 重新分配 choices, 直到两个 options 的 marginal return 相等。

Biological interpretation / failure condition. matching law 依 task reward schedule、choice stochasticity 与 learning assumptions。

Micro-check

Rescorla-Wagner error?

5.3 Rescorla-Wagner 与 blocking

【跨页连接】 cue weights 如何学习 immediate reward prediction?

$$V = \sum_i w_i x_i$$

$$\Delta w_i = \eta x_i (R - V)$$

1. squared prediction error 的 negative gradient 给 $x_i(R-V)$ 。
2. 未预测 reward 产生 positive error, active cue 增权。
3. 已由 cue S1 完全预测 reward 时, 再加入 S2 有 $R-V \approx 0$ 。
4. 因此 S2 没有学习信号, 产生 blocking。

Geometric/computational intuition. 学习由 surprise/prediction error 驱动, 而不是简单 stimulus-reward coincidence。

Biological interpretation / failure condition. configural representations、attention 与 non-stationary associations 会改变 blocking。

Micro-check

blocking 为什么发生?

5.4 TD error、temporal features 与 Q

【模型边界】 reward 延迟时如何把 credit 传回早期 cues/actions?

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$\delta_t = R_{t+1} + \gamma V_{t+1} - V_t$$

$$Q(s, a) = E[G_t \mid s_t = s, a_t = a]$$

1. γ 控制未来 reward discount。
2. TD error 比较当前 estimate 与 one-step reward+next value bootstrap target。
3. temporal/state features 使 cue 的不同 time-since-onset 有不同 weights。
4. Q 将 value 条件化于 action, 可支持 argmax action selection。

Geometric/computational intuition. prediction error 从 reward 时刻向 predictor cue 转移。
Biological interpretation / failure condition. partial observability、state representation 错误与 off-policy issues 在下一课系统化。

Micro-check

discounted return 中 $\gamma \rightarrow 0$ 表示什么?

6 Cross-page synthesis / 跨页因果与数学链

learned decisions 需要把 stimulus/action activity 与 delayed reward 联系起来。简单 reward-gated Hebb 在 reward 常出现时会持续增权；减去 expected reward baseline 后，更新变成 covariance rule。在特定 choice model/assumptions 下，stationary covariance condition 推出 matching behavior。Rescorla-Wagner 用 current reward prediction error 学 cue value；TD 将 error 延伸到 future value，dopamine 的 cue transfer/omission pattern 与 RPE 形式对应，但 dopamine 也可能混合 salience/movement signals。

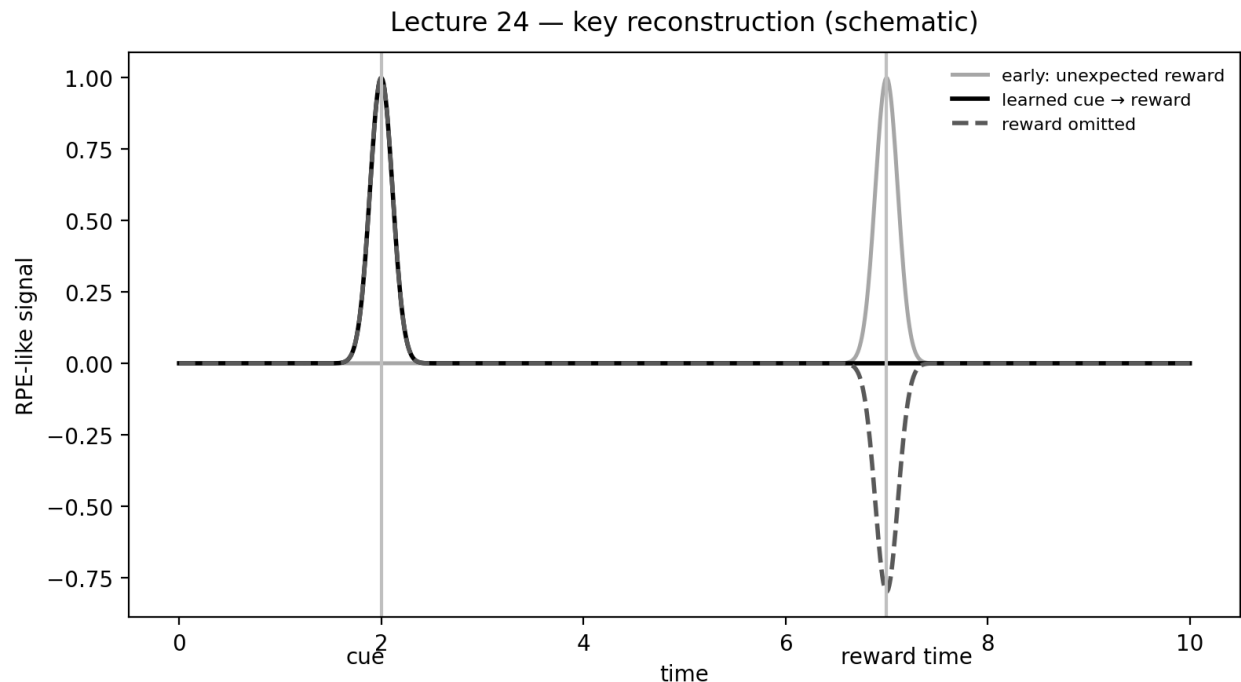


Figure note: schematic reconstruction; it encodes qualitative relations from the lesson and does not invent precise empirical data points.

7 Worked examples and transfer

7.1 dopamine RPE 三模式

【原课/核心例题】 Prompt. early unexpected reward、learned cue+expected reward、omitted reward。

Worked solution. early: reward 时 positive transient; learned: transient 转到 cue, reward 时近零; omission: cue positive、expected reward time negative dip。

Check. 这是 canonical RPE resemblance, 不排除 dopamine 同时编码 salience/movement。

7.2 TD 数值

【跨课连接】 Prompt. $R=0$, $\gamma=0.9$, $V_t=4$, $V_{t+1}=6$ 。

Worked solution. $\delta = 0 + 0.9 \times 6 - 4 = 1.4$, 当前 prediction 低, 应增加 active features' weights。
Check. next value 需 discount; 符号决定 potentiation/depression 方向。

Micro-check

Explain in your own words. 用不超过 120 字解释: 本课从 source page 1 到最后一页, 究竟把哪一个输入、状态、统计量或学习规则变成了可检验 prediction?

Self-reflection. 你最可能犯的是 concept、symbol、algebra、assumption 还是 causality error? 写出一个具体例子。

8 Formula and notation sheet

1. reward-baseline three-factor update

$$\Delta w = \eta e(R - \bar{R})$$

Conditions / conventions: eligibility e defined

2. covariance learning

$$E[\Delta w] = \eta \text{Cov}(e, R)$$

Conditions / conventions: stationary average

3. matching law

$$C_A/C_B \approx R_A/R_B$$

Conditions / conventions: specific foraging assumptions

4. linear value predictor

$$V = w^T x$$

Conditions / conventions: features x

5. Rescorla–Wagner update

$$\Delta w = \eta x(R - V)$$

Conditions / conventions: immediate/represented reward

6. TD/RPE

$$\delta = R + \gamma V_{next} - V$$

Conditions / conventions: Markov/state representation implicit

7. state-action value

$$Q(s, a) = E[G|s, a]$$

Conditions / conventions: policy/future behavior specified

9 Bilingual glossary

中文	English / symbol	Operational meaning
奖励学习 资格迹	reward learning eligibility trace	由 outcomes 调整 stimulus/action associations。 标记近期 pre/post/action activity 可被后续 reward 修改的变量。
奖励基线	reward baseline	expected reward, 用于计算 above/below-expectation signal。
匹配律	matching law	choice proportion 近匹配 obtained reward proportion。
价值	value	从当前 state/cue 预期的 discounted future reward。
Rescorla–Wagner 规则	Rescorla–Wagner rule	以 reward prediction error 更新 cue value。
阻断 时间差分误差	blocking temporal-difference error	已有 predictor 使新 cue 因零 error 而不学习。 reward+discounted next value–current value。

中文	English / symbol	Operational meaning
奖励预测误差	reward-prediction error, RPE	实际/更新 target 与 predicted value 的差。
Q 值	Q value	state-action conditioned expected return。

10 Common traps, assumptions, and limitations

1. reward-gated Hebb 不减 baseline, 导致所有 rewarded coactivity 持续增强。
2. matching 当作无条件理性最优 law。
3. blocking 解释成 S2 没有和 reward 同时出现。
4. 把 RPE、value 与 salience 混为一谈。
5. TD error 中漏掉 γ 或 next value。
6. 把 dopamine 视为所有 reward learning 的唯一第三因子。
7. 提前把本课简化 Q 概念当作完整 Q-learning algorithm。

Course-specific risk boundary. Lecture 24 是 biological/reward-learning bridge, Lecture 25 才系统化 MDP/RL。不要提前让 Bellman/Q-learning 淹没本课的 plasticity 与 dopamine 逻辑。

11 Cumulative Knowledge Check

Do not turn to the Hint Bank until you have written a first attempt.

1. naive reward-Hebb 为什么 over-strengthen?

2. 减 reward baseline 后 expected update 变什么?

3. matching 的 marginal-return condition?

4. matching law 的 ratio?

5. Rescorla–Wagner error?

6. blocking 为什么发生?

7. discounted return 中 $\gamma \rightarrow 0$ 表示什么?

8. TD error 公式?

9. cue learning 后 dopamine transient 为什么转移?

10. omission 时 negative dip 出现在哪里?

11. Q 与 V 区别?

12. third factor 可以是什么?

13. 公式表中的 “reward-baseline three-factor update” 解决什么问题? 写出适用条件与一个 sanity check。

14. 公式表中的 “covariance learning” 解决什么问题? 写出适用条件与一个 sanity check。

15. 公式表中的“matching law”解决什么问题？写出适用条件与一个 sanity check。

16. 公式表中的“linear value predictor”解决什么问题？写出适用条件与一个 sanity check。

17. 公式表中的“Rescorla–Wagner update”解决什么问题？写出适用条件与一个 sanity check。

18. 公式表中的“TD/RPE”解决什么问题？写出适用条件与一个 sanity check。

12 Hint Bank

1. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
2. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
3. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
4. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
5. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
6. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
7. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
8. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
9. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
10. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
11. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
12. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
13. **Hint 1.** 先从“reward-baseline three-factor update”对应的变量关系入手。
Hint 2. 把条件“eligibility e defined”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。
14. **Hint 1.** 先从“covariance learning”对应的变量关系入手。
Hint 2. 把条件“stationary average”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。

15. **Hint 1.** 先从 “matching law” 对应的变量关系入手。
Hint 2. 把条件 “specific foraging assumptions” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
16. **Hint 1.** 先从 “linear value predictor” 对应的变量关系入手。
Hint 2. 把条件 “features x” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
17. **Hint 1.** 先从 “Rescorla–Wagner update” 对应的变量关系入手。
Hint 2. 把条件 “immediate/represented reward” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
18. **Hint 1.** 先从 “TD/RPE” 对应的变量关系入手。
Hint 2. 把条件 “Markov/state representation implicit” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。

13 Complete Answer Key with reasoning

1. reward mean 为正时平均 update 持续为正。
2. eligibility 与 reward 的 covariance。
3. 两个 chosen options 的 expected reward rate 相等。
4. choice ratio $\approx$ reward ratio (在特定 assumptions 下)。
5. $R-V$ 。
6. 已有 cue 使 $V \approx R$, 新 cue 出现时 prediction error ≈ 0 。
7. 几乎只重视 immediate reward。
8. $\delta = R_{t+1} + \gamma V_{t+1} - V_t$ 。
9. cue 变成 earliest predictor, RPE 在 cue 时 positive, expected reward 时为零。
10. 预期 reward 应出现但未出现的时刻。
11. Q 条件化 action; V 只条件化 state/policy。
12. reward、RPE、error、contextual neuromodulator 等 gating signal。
13. 适用条件/约定: eligibility e defined。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\Delta w = \eta e(R - \bar{R})$$

14. 适用条件/约定: stationary average。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$E[\Delta w] = \eta \text{Cov}(e, R)$$

15. 适用条件/约定: specific foraging assumptions。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$C_A/C_B \approx R_A/R_B$$

16. 适用条件/约定: features x 。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$V = w^T x$$

17. 适用条件/约定: immediate/represented reward。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\Delta w = \eta x(R - V)$$

18. 适用条件/约定: Markov/state representation implicit。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\delta = R + \gamma V_{next} - V$$

14 Source-page concordance and coverage audit

Source file	Page	Coverage status
Lecture24-LearningFromRewards.pdf	1	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture24-LearningFromRewards.pdf	2	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture24-LearningFromRewards.pdf	3	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture24-LearningFromRewards.pdf	4	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture24-LearningFromRewards.pdf	5	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture24-LearningFromRewards.pdf	6	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.

15 Errata, uncertainty log, and external endnotes

15.1 Errata / source cautions

1. Page 1 将 dopamine 称 canonical candidate third factor; 本教材明确非唯一机制。
2. Page 3 的 matching derivation 依 choice-conditioned activity 与 reward-schedule assumptions.
3. Page 6 的 dopamine signals 与 RPE resemblance 是 influential interpretation, 不排除混合编码。

15.2 Uncertainty log

No unresolved handwriting uncertainty changes a core definition, equation, figure interpretation, or answer in this companion. Where handwriting was potentially ambiguous, the source-page image is preserved and the reconstruction avoids claiming unverified detail.

15.3 External endnotes

No external web or textbook source was used to replace the uploaded notes. Added material is limited to necessary mathematical scaffolding, explicit assumptions, standard sanity checks, and clearly labeled transfer examples.

15.4 Final self-audit

All source pages listed above were visually inspected. Equations were checked for symbol definitions, units/shapes, assumptions, algebraic transitions, limiting cases, and failure conditions. The PDF is static and contains no interactive form fields or scripts.